

Science Olympiad — Final Challenge Round 4

Names of participants: (Please print neatly)

School Name:

Team Number:

Check your team: ☐ Varsity ☐ JV1 ☐ JV2 ☐ JV3

Warning: Do not open this packet until given permission to do so.

Note: There are useful notes after this page.

Question	Value	Incorrect letters	Deduction	Score
1	0			
2	0			
3	0			
4	0			
Final Score				

1) [0 points] Decrypt the following cipher text that has been encoded using the Complete Columnar Cipher. The quote has **CHARA** somewhere in it. What does it say? INCLUDE THE X's AT THE END IF ANY WHEN CHECKING YOUR ANSWER

ESDCL CISYM NHIHE IEIET FNRAW LXHRE IAASA LIOTT

CTFRH PAEOA RTAM

Answer: _____

2) [0 points] Astronomers notice this repeating sequence in deep-space data. It's actually a quote, encoded with a Nihilist Substitution with a keyword length of 4. The deciphered text ends with **CHOSEPEACE**. What does it say?

60 74 47 58 38 83 65 69 66 63

65 85 66 72 76 65 39 62 46 59

46 65 44 75 58 92 76 58 67 65

56 75 39

	1	2	3	4	5
1					
2					
3					
4					
5					

E Q Q I D W I

[illegible]

EHFOZH FAGI PEAN UOXL

[illegible]